CS 6220: Project Proposal

The time has come to solve a data mining problem of your choice. You can work in teams of size **up to four**. The larger the team, the more options you have for choosing interesting problems. Of course, larger teams must deliver more results than smaller teams. In our experience you learn more in a team by discussing and collaborating with others, and it tends to be less work per person as data preparation and report writing can be shared. You may team up across the 2 sections of the course.

Each team must create all deliverables from scratch: it is not allowed to copy someone else's code or text, even if you modify it. If you use publicly available code/text, you need to indicate what was copied and cite the source in your report. The usual AI policy applies.

Key project events:

- Now: form teams and submit your project proposal.
- Final project reports will be due at the end of the semester.
- Each team presents their project in a video at the end of the semester.

This project proposal is worth 100 points and accounts for 2% of your final grade. (Final report and presentation account for 8% and 5%, respectively.) For late submissions you will lose one point per hour after the deadline. To encourage early work, you will receive a 10-point bonus if you submit your solution on or before the early submission deadline stated on Canvas. Notice that your total score cannot exceed 100 points, but the extra points would compensate for any deductions.

Project Requirements

There is no specific requirement about project scale, but the effort *per team member* should be comparable to fully solving both homework 2 and 3. You can pick any course-related problem, as long as you ensure the following:

- You perform class-related data mining tasks (prediction, pattern mining, clustering, and/or outlier detection).
- For *prediction*, you follow established practice, which usually means you train on training data, tune hyper-parameters on validation data, and determine model quality on test data. You also try to convince us that your final model neither significantly overfits nor underfits. Also determine if your model struggles on certain input groups, e.g., by grouping the test data in different ways on the input attributes and comparing per-group prediction quality.
- For *pattern mining*, analyze interestingness of association rules found. Discuss which patterns and rules could be dropped without losing "much in terms of interesting information".
- For *clustering*, use some of the cluster validity and quality measures. Also explore individual object pairs (same cluster, different clusters) and use your knowledge to determine if the clusters are meaningful.

- For *outlier detection*, explain clearly the intuition of "normal" vs "outlier" observations from the application point of view. Then argue why the specific outlier definition and technique you selected are the best match for that intuition.

The following examples illustrate typical project options:

- If there is a DM technique you want to understand in more depth, go ahead and implement it "from scratch".
- You want the right line in your resume: pick something that you believe will impress the recruiters and interviewers.
- You could just pick a problem from some existing competition, e.g., Kaggle.
- Is there a DM challenge to win some prize? Sure, try winning it using the project.
- Already have some results from a previous class or research project? Go ahead and expand them by using data mining.
- Excited about deep learning and LLMs? Download a free open-source LLM (maybe Llama or something else from Hugging Face) and use PyTorch or similar to fine-tune it for a specific problem you are interested in.
- Got a bunch of GPUs collecting dust? Figure out the maximum performance you can squeeze out using the right ML/DM library for some prediction problem of your choice.
- Spent too many years in school, got a job lined up, and this course stands between you and your vacation before starting the job? Well, you still need to deliver a good project. How about a standard prediction problem (MNIST handwritten digits?), where you explore several scikit-learn prediction techniques and tune their hyper parameters?

Project Proposal Report

Gradescope submission requirements:

- Select a "corresponding author" for your team. This is exactly 1 person.
- The corresponding author is the **only one** to submit the report for the team.
    - The corresponding author may overwrite a submitted report with a new version but never should multiple members of the same team submit additional reports.
- When submitting the report on Gradescope, the corresponding author must add all team members to the submission. (Otherwise the other team members will neither see our feedback nor receive any points.)

In your report, answer all the following questions:

1. State the names of all members of your team (at most four).
2. Show the link to your code repository, ideally on Github. Make sure instructor and TAs have access to it, but nobody else outside your team.
3. Which dataset(s) are you working with? You can choose any reasonable data.
    a. Make sure you have access to the data and download it immediately (if possible).

   b. Briefly explain the data. For example, “the dataset contains cloud-cover reports from hundreds of stations located on land and at sea. It is stored in CSV format. There are about 400 million records, each with 20 attributes describing properties such as latitude, longitude, time, and cloud cover at various altitudes. The dataset spans over 40 years.”
   c. Show a few records of the dataset. (Truncate long attribute values if needed.)
4. Explain the data mining problem you are trying to solve. Clarify the following:
   a. Type of problem, e.g., “predict label…” or “find clusters of … to understand …” or “find association rules like …” or “find outliers like…”.
   b. How do you plan to measure success?
   c. How do you plan to check if the results are meaningful? E.g., “group test data by … to compare prediction quality across groups” or “inspect individual test records that were (in-) correctly predicted to look for…” or “manually inspect pairs of objects in the same and different clusters” or “compare and contrast related association rules, looking for…”

Appendix: Links to Datasets

All datasets come **without any guarantee** that the links still work or the data is accessible. (They did work when we tried at some point in the recent past, but things on the Internet tend to change.) When choosing a dataset, make sure you can access it and that there is sufficient documentation to understand the structure and meaning of the data. It is also your responsibility to determine if there are any legal constraints that would prevent you from using the data for your project. **Inclusion of the links in this document does not mean that we endorse the use of these datasets in any way.**

Dataset	Location	Comments
Stanford Large Network Dataset Collection	https://snap.stanford.edu/data/	There are some huge graphs.
Variety of public data sets on Amazon’s AWS	http://aws.amazon.com/datasets/	
On-time performance of flights in the US, published by the Bureau of Transportation Statistics	http://www.transtats.bts.gov/DL_SelectFields.asp?Table_ID=236&DB_Short_Name=On-Time	
Various datasets, including Twitter and DBLP at UIUC	https://wiki.illinois.edu/wiki/display/forward/SoftwareDatasets	
Yelp Open Dataset	https://www.yelp.com/dataset	User reviews and information about businesses
Twitter dump (1 month)	https://wiki.cites.illinois.edu/wiki/display/for	This data collection contains actual tweets, i.e., the texts posted by users, not just follower information.

	ward/Dataset-UDI-TwitterCrawl-Aug2012	
Various datasets (you can search for topics like social science) at ICPSR (U Michigan)	https://www.icpsr.umich.edu/web/pages/ICPSR/index.html	
Million Song data set	http://labrosa.ee.columbia.edu/millionsong/	There are also a few complementary data sets with music ratings and song metadata.
Wikipedia data	http://en.wikipedia.org/wiki/Wikipedia:Database_download	
Wikimedia downloads	http://dumps.wikimedia.org/	
TPC-H benchmark	http://www.tpc.org/tpch/	You can create large benchmarks.
World Bank data	http://data.worldbank.org/	Many data sets are comparably small. Interesting Big Data challenges could arise from joining many of them.
Yahoo! Music ratings	http://webscope.sandbox.yahoo.com/catalog.php?datatype=c	
Other Yahoo! Data collections	http://webscope.sandbox.yahoo.com/catalog.php	
Stack Exchange Data Dump	http://blog.stackexchange.com/category/cc-wiki-dump/	There are many ways to use this for graph analysis, clustering, and prediction problems. It might be necessary to write code to convert from XML to a data format like CSV.
Netflix Prize data set	Not officially available any more.	
World-wide reports of cloud measurements	http://cdiac.ornl.gov/ftp/ndp026c/	
Open Library Project	https://openlibrary.org/data	There is data about editions of books and their authors.
Book reviews and ratings	http://www2.informatik.uni-freiburg.de/~cziegler/BX/	This data set could be used in combination with the above Open Library data.